

Daivya Shah

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EDUCATION

New York University, College of Arts & Science and Stern School of Business

New York, NY

B.A. in Computer Science and Data Science, Minor in Business

May 2026

- **GPA:** 3.64/4.00 | Junior/Senior Avg GPA: 3.72/4.00
- **Relevant Coursework:** Machine Learning, Deep Learning and Generative LLM Systems, Artificial Intelligence, Algorithms, Causal Inference, Data Management and Analysis, Probability and Statistics, Data Structures, Linear Algebra, Calculus III

EXPERIENCE

Newmark

Jun 2025 – Aug 2025

Software Engineer Intern

New York, NY

- Automated CapEx/OpEx forecasting for goal planning by building an internal OKR platform with **Golang** microservices and **FastAPI** on **Kubernetes (EKS)**, using **gRPC** for inter-service communication and surfacing **\$120K+** in projected annual labor cost savings
- Built a Real Estate Stacking Plan by standardizing **10K+** leasing records in **Databricks** using **Spark** and **Delta Lake**, enabling 300+ brokers to visualize floor-by-floor occupancy via a **Redis**-cached **React/TypeScript** interface, reducing query latency by **30%**
- Engineered a data ingestion pipeline with **Playwright**, **Apache Kafka**, and **LangChain** to extract and standardize data from **70K+** property listing URLs through multi-path scraping and LLM-based schema mapping, producing structured JSON for search and analytics

Harvest

Feb 2025 – Apr 2025

Founding Software/iOS Engineer

New York, NY

- Built and deployed an AI-powered manufacturer discovery platform in **Next.js** and **FastAPI**, with a native **SwiftUI** iOS app
- Scraped and normalized **100K+** manufacturer records from 500+ sites via **Selenium** into **Firebase**; built a hybrid semantic and keyword (**BM25**) search engine in **Pinecone** and a **Retrieval-Augmented Generation (RAG)** assistant for database-backed recommendations
- Trained a **LightGBM** learning-to-rank model with a LambdaRank objective on user behavior (clicks, saves) and manufacturer attributes (MOQ, region) to rank the best-fit manufacturers higher for each buyer search, improving top-5 ranking quality (NDCG@5) by **22%**

Everise

Jul 2024 – Dec 2024

Data Engineer Intern

New York, NY

- Developed a streaming pipeline to load customer call transcripts into **Snowflake** using **AWS Kinesis** and **Glue**, enabling NLP QA scoring models to automatically audit **78%** of a healthcare client's **50K+** monthly calls; visualized results in **Power BI** Quality Dashboard
- Extended the AWS Glue ingestion pipeline in **Python** with Protected Health Information (PHI) detection, automatically scanning and redacting 18 HIPAA identifier categories before loading to Snowflake, achieving zero compliance violations
- Built an **XGBoost** workforce demand forecasting model in Python using features engineered with **dbt** in Snowflake from Workday HCM and call detail records, reducing staffing error (MAPE) by **11%** and cutting WFM planners' manual reporting by 5+ hours weekly

eMeasurematics - Industrial Autonomous Solutions

May 2024 – Jul 2024

Machine Learning Intern

Chicago, IL

- Adapted a **computer vision**-based inventory system for a new steel plant by retraining a **YOLOv8** model on **8,400+** site-specific images using **OpenCV** and **SAM**-assisted labeling, improving coil and slab detection accuracy from **62%** to **91%** across 2 yard zones
- Built an **LSTM autoencoder** in **PyTorch** on **46M+** crane sensor records from 3 cranes over 6 months to detect mechanical issues missed by existing safety alerts, achieving **88%** precision and **83%** recall on 340 incidents with 38 ms inference latency
- Developed a **Grafana** dashboard on **InfluxDB** to visualize crane sensor data and flagged anomalies, helping engineers investigate issues

EXTRACURRICULAR ACTIVITIES & PROJECTS

NYU Machine Learning Club | Vice President

Jun 2024 – Present

- Organized speaker and hiring events with 10+ companies (Google, Meta, JPMorgan, etc.), growing membership to 1,100+ students

Ecole Polytechnique Urban Climate Hackathon - 5th Place/60 Teams

Nov 2025 – Dec 2025

- Built a climate analytics platform (**Flask**, **Vue.js**) with 26 REST API endpoints, enabling urban heat analysis across 12 cities
- Engineered a geospatial pipeline combining climate (ERA5), satellite vegetation, and weather station data for city-scale downscaling
- Trained **XGBoost**, **Random Forest**, **Ridge**, and **Quantile Regression** models across 186 hyperparameter trials for climate modeling
- Implemented **KMeans** clustering and **KNN** similarity analysis to identify climatically similar cities and shared temperature bias patterns

UCLA GenAI Hackathon - 1st Place/46 Teams

Jun 2024

- Headed the development of a **Generative Adversarial Network (GAN)** using **TensorFlow** to generate synthetic data, preserving the privacy of a dataset with **7M+** lines; achieved data replication of **99.87%**, ensuring high fidelity/accuracy of original data distributions
- Developed a data clean room solution, attesting to the confidentiality and integrity of the data by using **Intel TDX** based technologies

TECHNICAL SKILLS

Languages: Python, SQL, JavaScript, TypeScript, Golang, Java, Swift, HTML, CSS

Frameworks/Libraries: React, Node.js, FastAPI, Flask, PyTorch, TensorFlow, Scikit-learn, XGBoost, LangChain, PySpark, Pandas, NumPy

Technologies: AWS, Docker, Kubernetes, Apache Spark/Kafka, Databricks, Snowflake, PostgreSQL, MongoDB, Redis, Power BI, Grafana, Git

Concepts: System Design, CI/CD, Full Stack, OOP, Agile, Distributed Systems, ETL/ELT, Big Data, MLOps, Statistical Modeling